Perfect Subset Privacy in Polynomial Computation

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Motivation

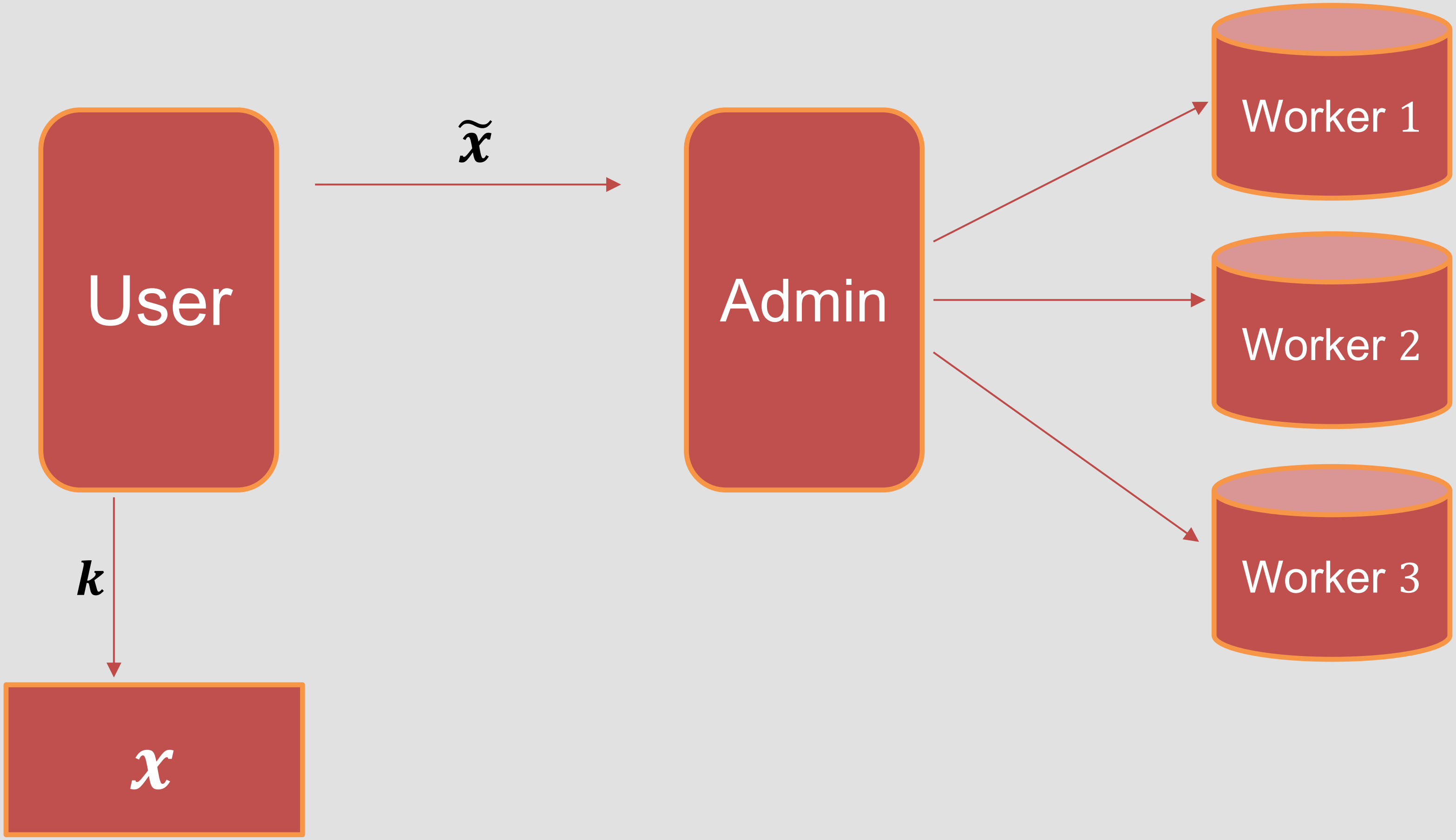
- Assigning large-scale computations to service providers raises privacy problems.
- Coded computing: application of error-correcting codes in distributed computations.
  - Used to address stragglers, adversaries, and privacy issues.
- Some level of restricted collusion is usually assumed among workers in the system.
  - The system administrator is trusted.

Motivation

- What about the setting where a user relies on a single <u>untrusted</u> service provider that introduces a privacy risk?
  - Very common nowadays (Amazon Web Services, Google Cloud, etc.)

- We seek to extend the scope of coded computation settings to protect privacy against the service provider as a whole.
  - The service provider has access to all of the data, which means assuming restricted collusion is not relevant.

Shifting the "Privacy Barrier"



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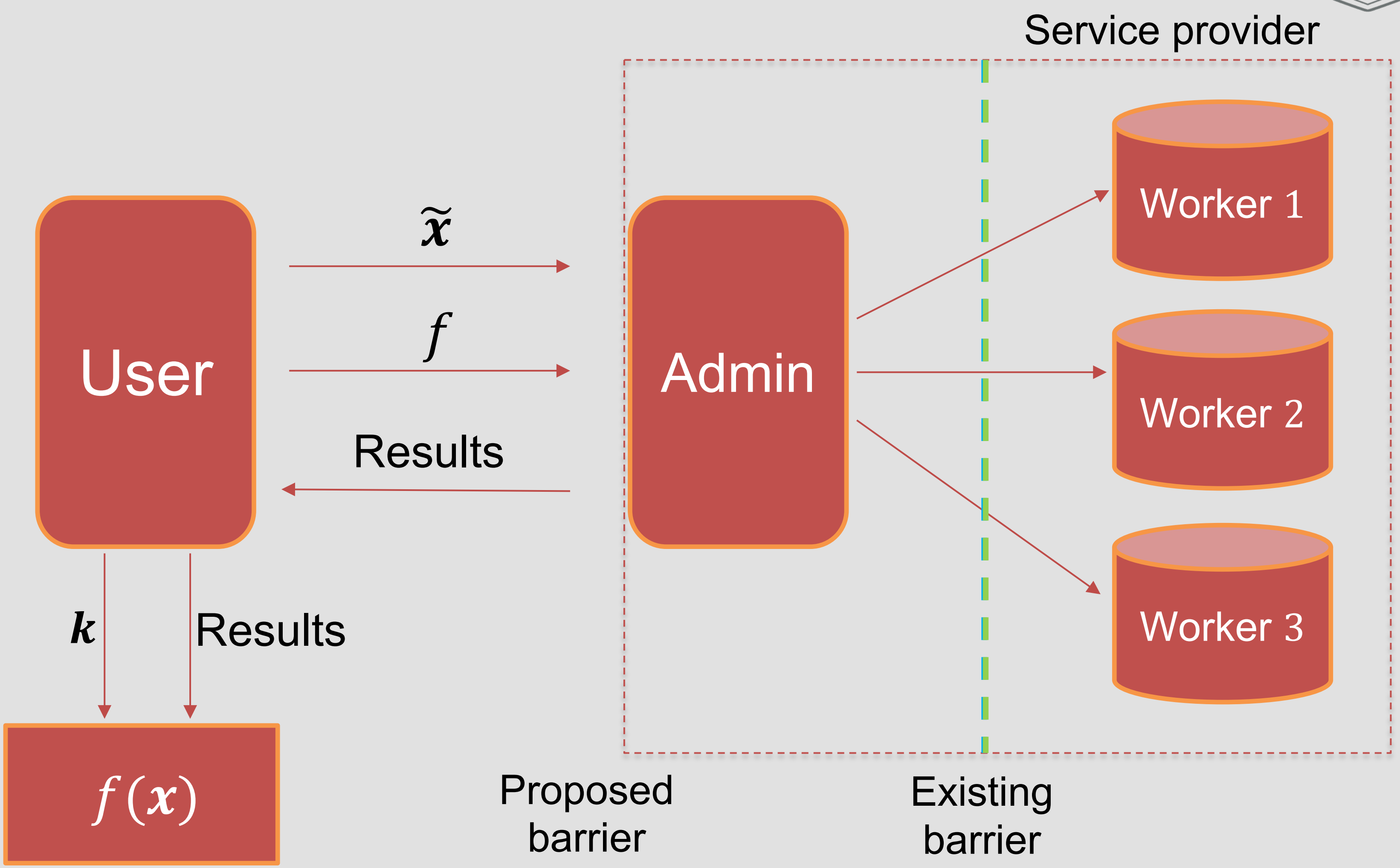
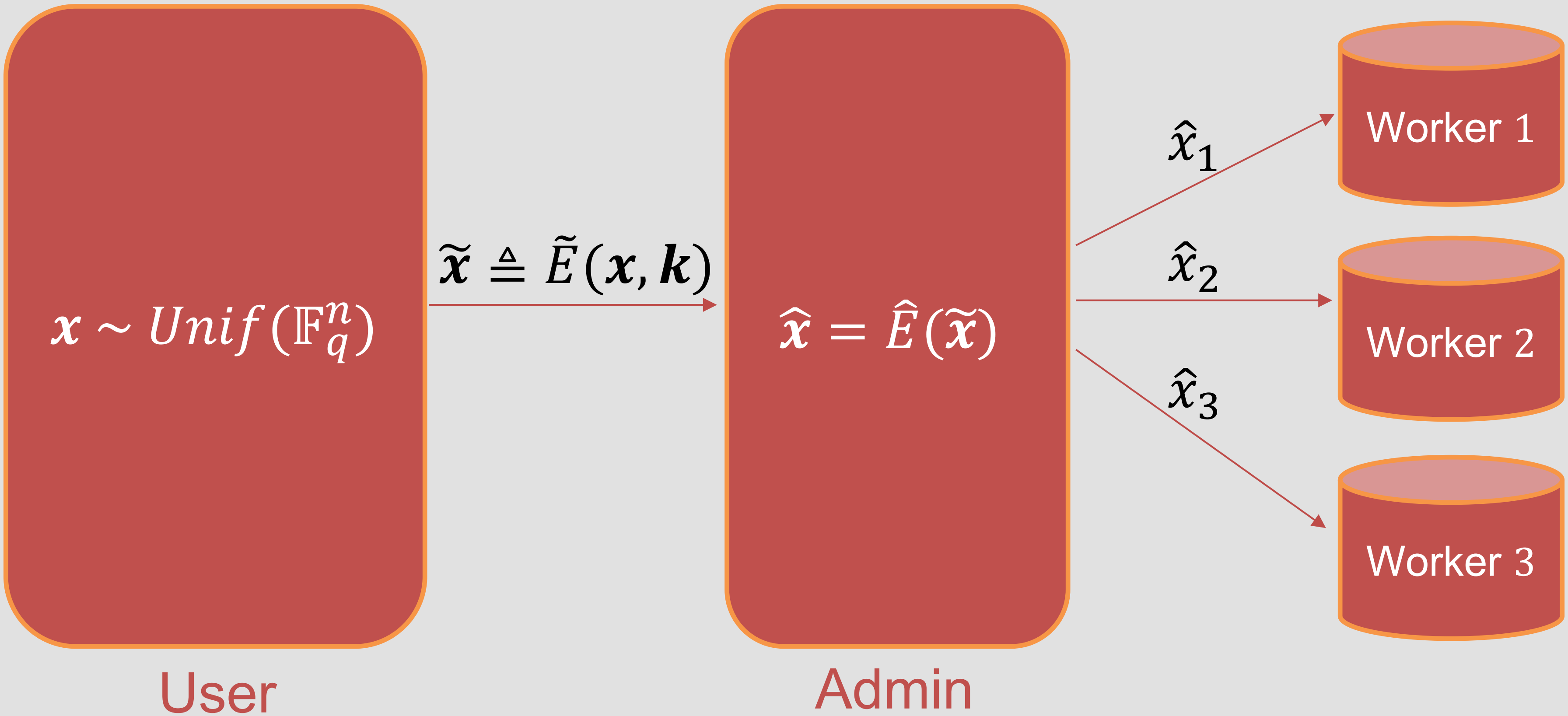
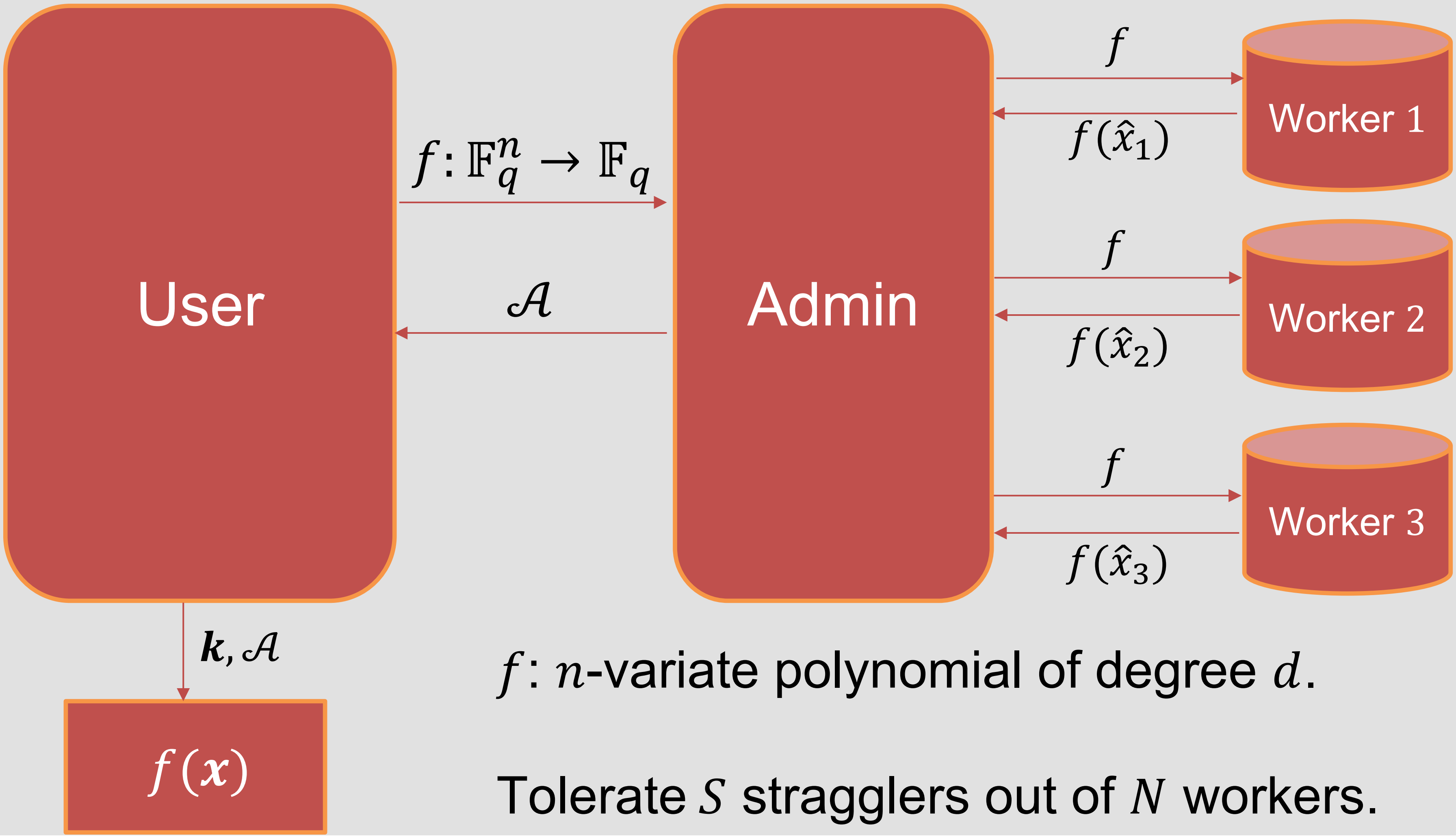


Computation Model: Storage



$\boldsymbol{k} \sim K = Unif(\mathbb{F}_q^m), m << n.$

Computation Model: Computation



f: n-variate polynomial of degree d.

Tolerate S stragglers out of N workers.

Perfect Subset Privacy

- Let $\tilde{X} \triangleq \tilde{E}(X, K) = (\tilde{X}_1, \ldots, \tilde{X}_n)$.
- An encoding scheme is *perfectly private* if $I(X; \tilde{X}) = 0$.
- Perfect privacy is a very strong notion.
  - No storage saving would be possible.
- Privacy guarantees of the form $I(X; \tilde{X}) \leq \epsilon$ provide no insight about which parts of the data are revealed.
- We leverage the notion of <u>perfect subset privacy</u>.

Perfect Subset Privacy

- <u>Definition</u>[1]: For a privacy parameter $r \in [n]$, an encoding procedure satisfies r*-subset privacy* if $I\left(X_{\mathcal{R}}; \tilde{X}\right) = 0$ for all subsets $\mathcal{R}$ of $[n]$ with $|\mathcal{R}| = r$, where $X_{\mathcal{R}} \triangleq (X_i)_{i \in \mathcal{R}}$.
- This notion of r-subset privacy provides uniform protection for all r-subsets of the data.
    - r serves as a tunable privacy parameter.
- <u>Our goal</u>: devise an encoding method that achieves perfect subset privacy.

[1] N. Raviv and Z. Goldfeld, "Perfect subset privacy for data sharing and learning," in 2022 IEEE International Symposium on Information Theory (ISIT), IEEE, 2022, pp. 1850–1855.

Performance Metrics

- We judge the merit of schemes by:
  - Number of workers needed, measured by N;
  - Download cost, measured by the number D of $\mathbb{F}_q$ symbols downloaded by the user;
  - Side information, measured by m, the size of random key $\boldsymbol{k}$.
- Ideally, we would like to minimize all three quantities.

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Achieving r-subset Privacy

- [1]Raviv et al. presented a method achieving perfect subset privacy when the field size is at least the length of data.
- <u>Our approach</u> (a generalization):
  - Let $\tilde{\boldsymbol{x}} \triangleq \boldsymbol{x} + \boldsymbol{k}G,$ where:
  - $\boldsymbol{k} \sim Unif\left(\mathbb{F}_q^m\right),$ and
  - $G \in \mathbb{F}_q^{m \times n}$ is a generator matrix for any $[n, m]_q$ linear code $\mathcal{C}$ that satisfies $d_{min}(\mathcal{C}^{\perp}) \geq r + 1$.
- Notice $m \geq r$ by the Singleton bound, with equality achieved with MDS codes.

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Trivial Scheme

- A trivial computation scheme has download cost q^m.
  - The user sends $\tilde{\boldsymbol{x}} \triangleq \boldsymbol{x} + \boldsymbol{k}G$ to the admin.
  - The admin encodes $\tilde{\boldsymbol{x}}$ to $(\tilde{\boldsymbol{x}} - \boldsymbol{t}G \colon \boldsymbol{t} \in \mathbb{F}_q^m)$.
  - Later, the admin gets f from the user and has workers compute $\mathcal{A} \triangleq (f(\tilde{\boldsymbol{x}} - \boldsymbol{t}G) \colon \boldsymbol{t} \in \mathbb{F}_q^m)$.
  - The user retrieves $f(\boldsymbol{x}) = f(\tilde{\boldsymbol{x}} - \boldsymbol{k}G)$.

Information Sets for Reed-Muller Codes

- We aim to reduce download cost via *information sets for Reed-Muller codes*.
- Given an $[n, k]$ linear code $\mathcal{C}$, an k-element subset of $[n]$ is an *information set* for $\mathcal{C}$ if the corresponding k columns of the generator matrix for $\mathcal{C}$ is linearly independent.
- Let $RM_q(d, m)$ be the Reed-Muller code over $\mathbb{F}_q$, where:
  - $d =$ total degree.
  - $m =$ # variables.
  - Let $\mathfrak{I}$ be an information set for $RM_q(d, m)$.

Lowering Download Cost

- The user encodes $\boldsymbol{x}$ to $\widetilde{\boldsymbol{x}} \triangleq \boldsymbol{x} + \boldsymbol{k}G$ as defined previously. Then, the system admin encodes $\widetilde{\boldsymbol{x}}$ to $(\widetilde{\boldsymbol{x}} - \boldsymbol{t}G : \boldsymbol{t} \in \boldsymbol{\mathfrak{T}})$ and distributes them to workers.
- Later, the admin receives from the user polynomial f of total degree d.
- Define polynomial $g: \mathbb{F}_q^m \to \mathbb{F}_q$ as $g(\boldsymbol{t}) \triangleq f(\widetilde{\boldsymbol{x}} - \boldsymbol{t}G)$ for all $\boldsymbol{t} \in \mathbb{F}_q^m$.
  - Total degree of g is at most d.
  - $g(\boldsymbol{k}) = f(\boldsymbol{x})$.

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- Later, the admin receives from the user polynomial f of total degree d.
- Define polynomial $g: \mathbb{F}_q^m \to \mathbb{F}_q$ as $g(\boldsymbol{t}) \triangleq f(\widetilde{\boldsymbol{x}} - \boldsymbol{t}G)$ for all $\boldsymbol{t} \in \mathbb{F}_q^m$.
- The admin obtains $\mathcal{A} \triangleq (f(\widetilde{\boldsymbol{x}} - \boldsymbol{t}G): \boldsymbol{t} \in \mathfrak{T})$ from the workers and sends it to the user.
- The user can evaluate g at all points and compute the desired $g(\boldsymbol{k}) = f(\boldsymbol{x})$ from $\mathcal{A}$ and the key $\boldsymbol{k}$.

Lowered Download Cost

- Theorem: The information set scheme requires download cost $D = |\mathfrak{T}| = \lambda$, where λ is the dimension of $RM_q(d, m)$.
- Example: Take the dual code $C^{\perp}$ as the $\left[n = 2^t - 1, k \geq n - \frac{r}{2}t, d_{min} \geq r + 1\right]$ binary BCH code, where $t \geq 3$ is an integer and r is an even positive integer.
  - $D \leq \left(1 + \frac{r}{2}\log(n+1)\right)^d = poly(\log(n))$, assuming d is a constant and $r = O(poly(\log(n)))$.

Performance Metrics

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Straggler Resilience

- Repetition approach requires $N = (S+1)D$ workers.
- Lagrange Code Computing (LCC)-based approach requires $N = (\lambda - 1)d + S + 1$ workers.
  - Requires $q \geq N$.
- Notice that $(\lambda - 1)d + S + 1 < (S+1)\lambda$ if $d < S$.
- Method of *information super-sets*: no field size restriction.

Straggler Resilience via Information Super-sets

- Definition: Let $\mathcal{C}$ be a linear code. A multiset $\mathcal{T} \subseteq [n]$ is called an S-*information super-set* for $\mathcal{C}$ if every $(|\mathcal{T}| - S)$ subset of $\mathcal{T}$ contains an information set for $\mathcal{C}$.
- Let $g(\boldsymbol{t}) \triangleq f(\widetilde{\boldsymbol{x}} - \boldsymbol{t}G)$ for all $\boldsymbol{t} \in \mathbb{F}_q^m$ and $\widetilde{\boldsymbol{x}} \triangleq \boldsymbol{x} + \boldsymbol{k}G$ as before and let $\mathcal{T}$ be an S-information super-set for $RM_q(d,m)$.
- The admin encodes $\widetilde{\boldsymbol{x}}$ to $(\widetilde{\boldsymbol{x}} - \boldsymbol{t}G : \boldsymbol{t} \in \boldsymbol{\mathcal{T}})$ and distributes to workers.
- By definition, the admin is able to receive evaluations of g at an information set for $RM_q(d,m)$ even with S stragglers.
- $N = |\mathcal{T}|$.

Explicit Information Super-sets

- Explicit construction of information super-sets is interesting.
  - Some results are implied by puncturing Reed-Muller codes[1].
- We have successfully come up with constructions of:
  - 1-information super-sets for $RM_2(1, m)$ of size $m + 2$ (m even) and $m + 3$ (m odd);
  - 2-information super-sets for $RM_2(1, m)$ of size $2m + 1$ (smaller sizes possible via greedy search);
  - S-information super-sets for $RM_2(d, m)$ using the $(u, u + v)$-construction of binary Reed-Muller codes.

[1] V. Guruswami, L. Jin, and C. Xing, "Efficiently list-decodable punctured reed-muller codes," IEEE Transactions on Information Theory, vol. 63, no. 7, pp. 4317–4324, 2017.

Summary

- This work shifts the focus of privacy in coded computing to safeguarding the privacy of the user against the service provider as a whole.
- We leverage perfect subset privacy, which guarantees zero information leakage from all subsets of the data up to a certain size.
- Using techniques from Reed-Muller decoding, we provide schemes which enable polynomial computation with perfect subset privacy in straggler-resilient systems.

Thank you!